



Artificial Intelligence Foundations (CIS4049-N)

Module Introduction

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Teaching Team



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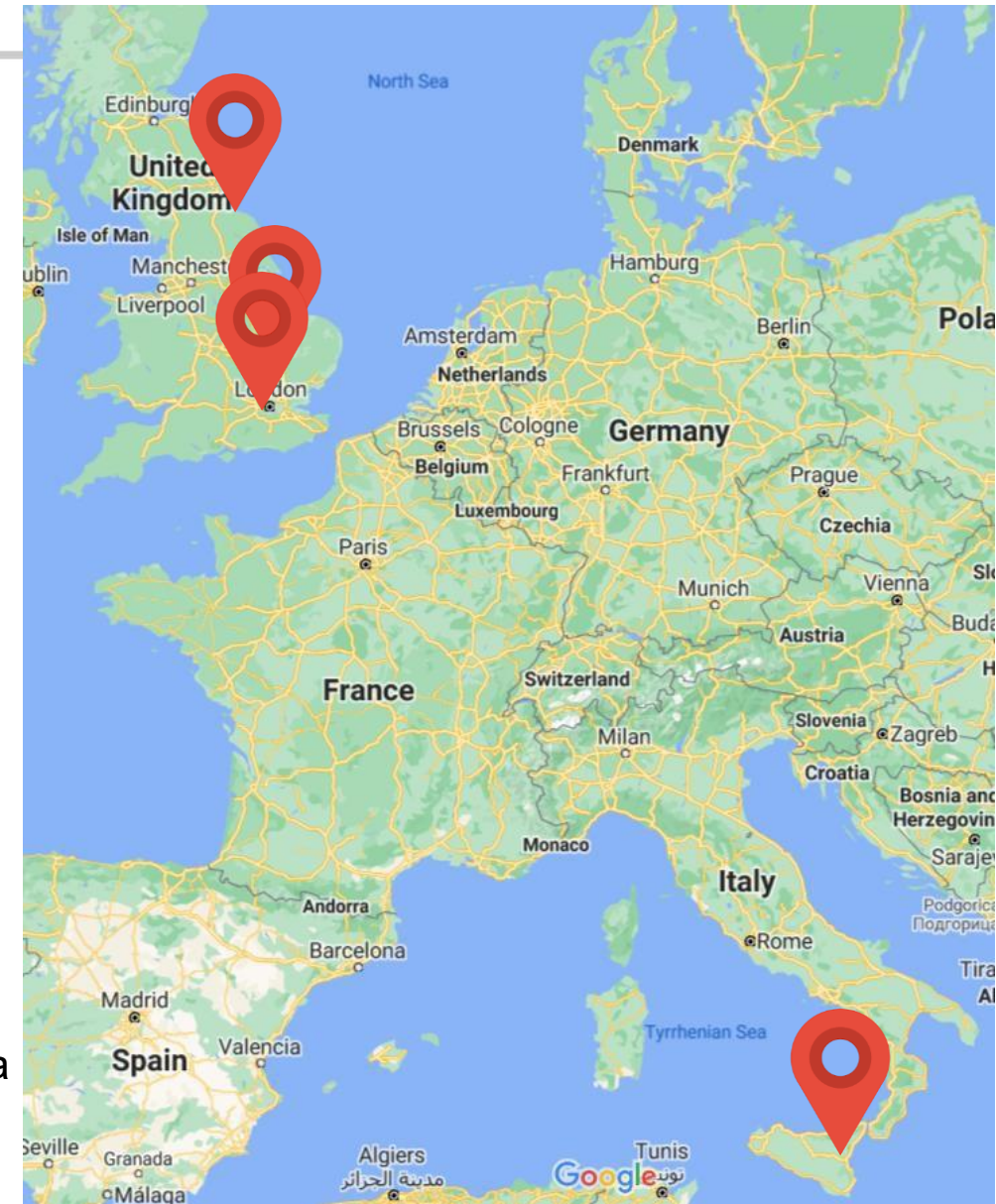


My research interests include **game theory**, **network science**, and **artificial intelligence/machine learning**. For further information about my research profile, please visit the Teesside University research portal or my GitHub web page: <https://distefanoalessandro.github.io>



About me

- **Senior Lecturer in Computer Science** at Teesside University
- **Research Associate** at Computer Laboratory, Department of Computer Science and Technology, University of Cambridge, UK, supervised by **Prof. Pietro Liò**, Full Professor and member of the AI group (by the end of August)
- **Research Associate / Teaching Assistant** at King's College London, Department of Engineering, UK - Course "Network Theory" - **Dr. Nishanth Sastry**, Senior Lecturer, Department of Engineering, King's College, London, UK (now full professor Joint Head of the Distributed and Networked Systems Group at Department of Computer Science, University of Surrey)
- **Postdoctoral researcher** at DIEEI, University of Catania (2016-2019)
- **PhD in Systems Engineering** at Department of Electrical, Electronic and Software Engineering (DIEEI), University of Catania, Italy. PhD Thesis: *"Evolutionary Dynamics of Social Behaviours on Multilayer Networks"* (2015)
- **BSc and MSc in Telecommunications Engineering** at University of Catania
- **Collaborations also with other universities** (Missouri University S&T, etc.)



Teaching Team



- **Dr Chaimaa Tarzi, Lecturer in Computing & Games, SCEDT** – Email: C.Tarzi@tees.ac.uk
- **Dr Shatha Ghareeb, Lecturer in Computer Science, SCEDT** – Email: S.Ghareeb@tees.ac.uk
- **Dr Hiba Alsmadi, Lecturer in Data Visualization, SCEDT** – Email: H.Alsmedi@tees.ac.uk
- **Dr Aliyu Abubakar, Lecturer in Computer Science, SCEDT** – Email: A.Abubakar@tees.ac.uk
- **Suraj Verma – Part-time Lecturer in Computer Science /PhD student (final year)** – Email: S.Verma@tees.ac.uk
- **Le Minh Thao Doan – Part-time Lecturer in Computer Science/PhD student (4th year)** – Email: L.Doan@tees.ac.uk
- **Kaveh Shahhosseini – Part-time Lecturer in Computer Science/PhD student (3rd year)** – Email: K.Shahhosseini@tees.ac.uk
- **Andrew Powell – Part-time Lecturer in Computer Science/PhD student (1st year)** – Email: A.Powell@tees.ac.uk

See Blackboard for online materials: <https://bb.tees.ac.uk/>

Artificial Intelligence Foundations

Module **Aims:**

- provide an introduction to Artificial Intelligence (AI)
- introduce AI concepts, applications and theory
- explore the key factors that have made the AI successful for various applications
- give an awareness of the current state and direction of AI research

Approach to Teaching

Lectures/Seminars (1 Lecture – 1 hour)	• presentation of concepts and theories • work and practice on AI examples to embed knowledge
Lab Sessions (2 hours – once a week)	• AI solutions design, exercises and application to real-world cases • Implementation of AI notions and techniques

Teaching Approach and Lectures Schedules



- Lecture: **Tuesday 11 am – 12 pm**

Room: **OL1 (Europa/IT building – ground floor) – Dr Alessandro Di Stefano**

- Lab sessions: Look at your timetable
- Each week will consist of **one lecture** (1 hour) and a **practical session** (2 hours); you will be required to do work outside the class-time
- You must **attend** (listen) all **lectures** and **labs**, and ensure you complete the practical assignment every week by working between class sessions
- This module is fundamental to your career and for your life!

Learning Outcomes

Personal and Transferable Skills (PTS)

1. Effectively communicate and evaluate complex information related to AI theory
2. Use personal reflection to analyse self and own actions in the course of working as a group or individual in a real-world AI case study
3. Reflect upon, take ownership of and critically appraise the outcome of an implemented solution against a given brief for a simulated or real-world problem using appropriate AI technologies

Research, Knowledge and Cognitive Skills (RKC)

4. Critically analyse a solution to a given real-world case study
5. Utilise effectively AI software and techniques in problem solving
6. Critically appraise of recent scientific literature in AI in the context of a given scenario

Professional Skills (PS)

7. Critically evaluate the commercial risks and opportunities related to solving an AI problem
8. Autonomously evaluate improvements to performance drawing on innovative or best practice in applied AI related skills

Assessment

ICA (100%): one in-course assessment

Written report (equivalent to 4,000 words), an **artefact** (AI solution), and a brief **walk-through video**, will assess the **learning outcomes**: evaluating and reflecting on the application of AI techniques to a real-world case study and justify the use of these techniques.

ICA Briefing – Explain the details of the ICA

Topics covered

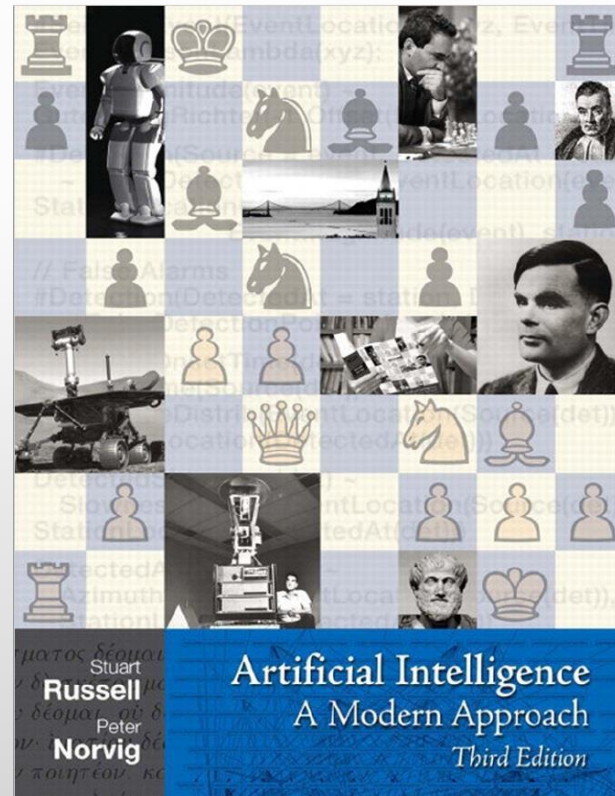
- What's AI?
- Intelligent agents
- Problem-Solving:
 - Searching techniques: Uninformed and Informed Search methods
- Reinforcement Learning
- Large Language Models (LLMs)

Schedule



Week	Lecture	Practical/Seminar/Tutorial/Lab
1	What's AI	Introduction to Python
2	Intelligent Agents & Pillars of AI	Python basics / Programming Agents with Python
3	Uninformed Search I	Uninformed Searching Algorithms - Implementation with R and Python (BFS, DFS)
4	Uninformed Search II	Uninformed Searching Algorithms - Implementation with R and Python (BFS, DFS)
5	Informed Search	Informed Search Strategies – Implementation with R and Python (Greedy, A*, Dijkstra) - ICA examples
6	Introduction to LLMs	LLMs & Prompting
7	LLMs – Part II	Working with LLMs
8	LLMs – Part III	Search & LLMs - ICA examples ICA Progress & Feedback
9	Reinforcement Learning	Reinforcement Learning with R / Reinforcement Learning with Python
10	Reinforcement Learning – Applications	Reinforcement Learning – ICA examples
11	Assessment/Consolidation	ICA Support / Students' presentation
12	Assessment/Consolidation	ICA Support / Students' presentation

Russell, S.J. and Norvig, P., (2019). Artificial intelligence: a modern approach. Pearson 4th Edition. ISBN-10: 0134610997



What do you need to do?

- **Ask questions** (please write me if you need any clarification)
- **Interact** and **be active** in your learning
- Take **responsibility** for your learning
- Solve all the **exercises / quizzes** (lectures and labs)
- Do **extra work** (read recommended books or readings, etc.)

Questions?

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